

4 (long)

(a) Figure 4(a) shows the circuit for a general AC bridge, in which the voltage source is a sine wave with angular frequency ω . Define what is meant by the balance condition for this circuit, and derive the condition for balance in terms of Z_1 to Z_4 . [5]

(b) Using the result of part (a), derive an expression for the frequency at which the AC bridge shown in Fig. 4(b) is balanced. Then derive the relationship between the other components. [13]

(c) In a similar manner, derive the balance conditions for the AC bridge as shown in Fig. 4(c) in terms of its component values and ω . [12]

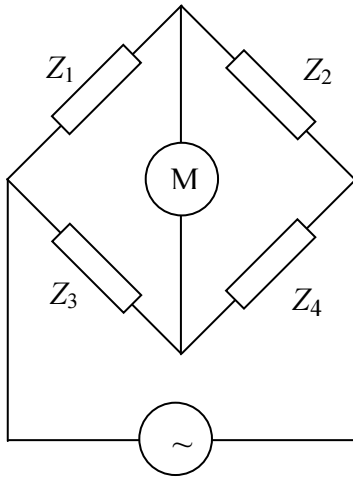


Fig. 4 (a)

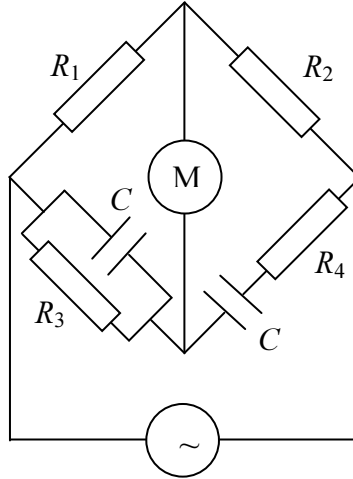


Fig. 4(b)

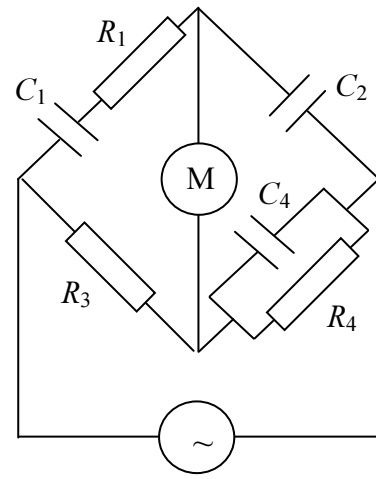


Fig. 4(c)